



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Team ID	FNO2			
Semester	6 th			
Course :	Interdisciplinary Project			
Team Details		Program	USN	Name
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	2.	AIML	1RV23AI035	Gnanendra Naidu N
	3.	ECE	1RV23EC023	Arumita Nandi
	4.	ECE	1RV23EC183	Vinay A
Project Title	FoodSense: Multimodal AI System for Early Spoilage Prediction			
Center of Excellence	NA			
Internal Guide				
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Designation &	Assistant Professor			
Department	Department of Computer Science Engineering			

INTRODUCTION:

Food spoilage is a major problem in households, restaurants, and food storage systems. Improper storage conditions such as high temperature, excess humidity, and bacterial growth can cause food to spoil quickly, leading to health risks and increased food waste. Traditional food containers do not provide any information about the condition or freshness of the food stored inside them.

OBJECTIVES:

- **Monitor Food Freshness in Real Time:**
Develop a system that continuously tracks factors such as temperature, humidity, and gas levels inside the container to determine the freshness and safety of stored food.
- **Provide Alerts for Food Spoilage:**
Design the container to notify users through indicators or a connected mobile application when food is close to spoiling or when storage conditions become unsafe.
- **Reduce Food Waste and Improve Storage Efficiency:**
Enable smarter food management by helping users monitor storage conditions and consumption timelines, thereby minimizing unnecessary food waste and maintaining food quality.

METHODOLOGY:



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SOFTWARE REQUIREMENTS:

- OpenCV
- Arduino IDE
- Pytorch

HARDWARE REQUIREMENTS:

- Microcontroller (e.g., Arduino / ESP32)
- Temperature Sensor (e.g., DHT11/DHT22)
- Humidity Sensor
- Gas Sensor (e.g., MQ series)

INTERDISCIPLINARY RELEVANCE:

• Role of ECE Students:

ECE students focus on the hardware development of the system, working on integrating sensors, the ESP32, and communication modules including circuit design, sensor interfacing, power management, and ensuring reliable data transmission from the container.

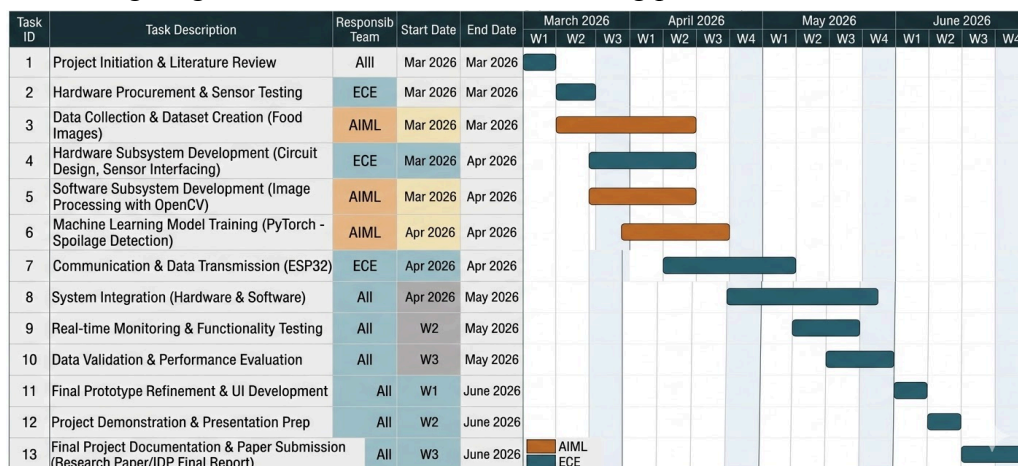
• Role of AIML Students:

AIML students focus on the software and intelligent analysis part of the project. Using OpenCV and Python, they develop algorithms to analyze images of food and detect signs of spoilage such as color change or mold formation.

INNOVATION TO THE FIELD:

- **Multimodal sensing:** Combines gas sensors, moisture sensor (VPD monitoring), temperature sensors, and camera-based computer vision to detect early spoilage signals.
- **Advanced CV algorithms:** Uses Vision Transformer-based image analysis to capture subtle color, texture, and surface changes in food.
- **Temporal sensor modeling:** Gas and moisture changes over time are analyzed to detect early biochemical spoilage patterns.
- **Explainable AI integration:** Techniques like Grad-CAM, LIME, and saliency maps highlight important image regions or sensor features influencing predictions.

TIMELINE (GANTT CHART):



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Designation Department

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